



LEADING EDGE INTERNATIONAL
AVIATION ACADEMY INC.

RADIOTELEPHONY



OBJECTIVE

To inform students of radiotelephony procedures and phraseology as applied to aircraft operations under IFR, and actions to be taken in case of communication failure.

OVERVIEW

- General Definitions and Phraseologies
- General Operating Procedures
 - ◆ Transmitting Technique
 - ◆ Position Reporting
 - ◆ Change from IFR to VFR Flight
 - ◆ International Phonetic Chart
 - ◆ Read Back
 - ◆ Radio Check
- Radio communications for general operating procedures
 - ◆ Different Communication Facilities
 - ◆ Air Traffic Control Clearances
 - ◆ Instrument Flight Sequence
 - Procedures for Departing Aircraft
 - Procedures for Arriving Aircraft
- Action required to be taken in case of emergencies
 - ◆ Communication failure
 - ◆ Distress and urgency procedures
 - ◆ Emergency Frequencies
 - ◆ Squawk Codes
 - ◆ Emergency message procedure
 - ◆ International Emergency signals
 - ◆ 4 C's of an emergency
 - ◆ ATC light gun signals
- General principles of VHF propagation and allocation of frequencies
- Morse code

GENERAL DEFINITIONS

Manual of Standards for Air Traffic Services

- **Aerodrome.** A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft.
- **Air traffic control clearance.** Authorization for an aircraft to proceed under conditions specified by an air traffic control unit.
 - Note 1. — For convenience, the term “air traffic control clearance” is frequently abbreviated to “clearance” when used in appropriate contexts.
 - Note 2. — The abbreviated term “clearance” may be prefixed by the words “taxi”, “take-off”, “departure”, “en-route”, “approach” or “landing” to indicate the particular portion of flight to which the air traffic control clearance relates.
- **Airway.** A control area or portion thereof established in the form of a corridor.
- **Area navigation (RNAV).** A method of navigation which permits aircraft operation on any desired flight path within the coverage of ground- or space-based navigation aids or within the limits of the capability of self-contained aids, or a combination of these.
- **Automatic terminal information service (ATIS).** The automatic provision of current, routine information to arriving and departing aircraft throughout 24 hours or a specified portion thereof.

GENERAL DEFINITIONS

Manual of Standards for Air Traffic Services

- **Blind transmission.** A transmission from one station to another station in circumstances where two-way communication cannot be established but where it is believed that the called station is able to receive the transmission.
- **Controlled airspace.** An airspace of defined dimensions within which air traffic control service is provided in accordance with the airspace classification.
- **Final approach.** That part of an instrument approach procedure which commences at the specified final approach fix or point, or where such a fix or point is not specified,
 - a) at the end of the last procedure turn, base turn or inbound turn of a racetrack procedure, if specified; or
 - b) at the point of interception of the last track specified in the approach procedure; and ends at a point in the vicinity of an aerodrome from which: a landing can be made; or a missed approach procedure is initiated.
- **Flight Service Station.** An aeronautical telecommunication facility that provides any or all of the following: aerodrome advisory service, aerodrome flight information service, pre-flight briefing service, flight following service, enroute air- ground or in-flight assistance service and alerting service.

Note. – Flight assistance service may include aerodrome advisory service, pre-flight briefing service, flight following service, enroute air-ground or in-flight assistance service and alerting service.

- **Holding fix.** A geographical location that serves as a reference for a holding procedure.
- **Holding procedure.** A predetermined maneuver which keeps an aircraft within a specified airspace while awaiting further clearance



GENERAL DEFINITIONS

Manual of Standards for Air Traffic Services

- **Initial approach segment.** That segment of an instrument approach procedure between the initial approach fix and the intermediate approach fix or, where applicable, the final approach fix or point.
- **Instrument approach procedure (IAP).** A series of predetermined maneuvers by reference to flight instruments with specified protection from obstacles from the initial approach fix, or where applicable, from the beginning of a defined arrival route to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or en-route obstacle clearance criteria apply.
- **Instrument meteorological conditions (IMC).** Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, less than the minima specified for visual meteorological conditions.
- **Missed approach procedure.** The procedure to be followed if the approach cannot be continued.
- **Procedure turn.** A maneuver in which a turn is made away from a designated track followed by a turn in the opposite direction to permit the aircraft to intercept and proceed along the reciprocal of the designated track.
- **Reporting point.** A specified geographical location in relation to which the position of an aircraft can be reported.

GENERAL DEFINITIONS

Manual of Standards for Air Traffic Services

- **Runway incursion.** Any occurrence at an aerodrome involving the incorrect presence of an aircraft, vehicle or person on the protected area of a surface designated for the landing and take-off of aircraft.
- **Runway visual range (RVR).** The range over which the pilot of an aircraft on the center line of a runway can see the runway surface markings or the lights delineating the runway or identifying its center line.
- **Significant point.** A specified geographical location used in defining an ATS route or the flight path of an aircraft and for other navigation and ATS purposes. Note. — There are three categories of significant points: ground-based navigation aid, intersection and waypoint. In the context of this definition, intersection is a significant point expressed as radials, bearings and/or distances from ground-based navigation aids.
- **Standard instrument arrival (STAR).** A designated instrument flight rule (IFR) arrival route linking a significant point, normally on an ATS route, with a point from which a published instrument approach procedure can be commenced.
- **Standard instrument departure (SID).** A designated instrument flight rule (IFR) departure route linking the aerodrome or a specified runway of the aerodrome with a specified significant point, normally on a designated ATS route, at which the en-route phase of a flight commences.
- **Terminal control area (TMA).** A control area normally established at the confluence of ATS routes in the vicinity of one or more major aerodromes

GENERAL DEFINITIONS

Manual of Standards for Air Traffic Services

- **Traffic information.** Information issued by an air traffic services unit to alert a pilot to other known or observed air traffic which may be in proximity to the position or intended route of flight and to help the pilot avoid a collision.
- **Transition altitude.** The altitude at or below which the vertical position of an aircraft is controlled by reference to altitudes.
- **Transition level.** The lowest flight level available for use above the transition altitude.
- **Visual approach.** An approach by an IFR flight when either part or all of an instrument approach procedure is not completed and the approach is executed in visual reference to terrain.
- **Waypoint.** A specified geographical location used to define an area navigation route or the flight path of an aircraft employing area navigation. Waypoints are identified as either: Fly-by waypoint. A waypoint which requires turn anticipation to allow tangential interception of the next segment of a route or procedure, or Flyover waypoint. A waypoint at which a turn is initiated in order to join the next segment of a route or procedure.

GENERAL OPERATING PROCEDURES

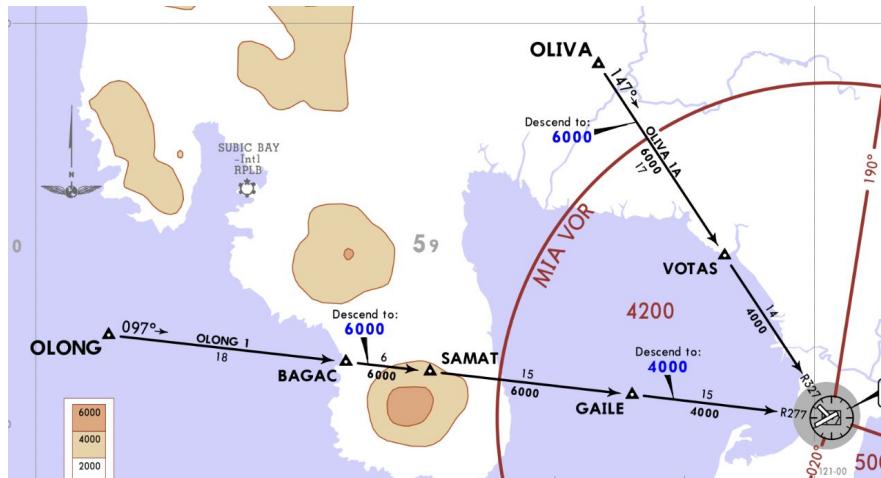
TRANSMITTING TECHNIQUE

- Before transmitting, listen on the frequency to be used;
- Be familiar with good microphone techniques;
- Use a normal conversation tone;
- Maintain an even rate of speech not exceeding 100 words per minute;
- Maintain speaking volume at a constant level;
- Slightly pause before and after numbers;
- Avoid using hesitant sounds;
- Suspend speech when turning head away from mic;
- Depress transmit switch fully when message is complete;
- Long messages should be interrupted momentarily.

GENERAL OPERATING PROCEDURES

POSITION REPORTING

On routes defined by designated significant points, position reports shall be made by the aircraft when over, or as soon as possible after passing, each designated compulsory reporting point. Additional reports over other points may also be requested by the ATS unit.



GENERAL OPERATING PROCEDURES

POSITION REPORTING

On routes not defined by designated significant points, position reports shall be made by the aircraft as soon as possible after the first half hour of flight and at hourly intervals thereafter, except under *conditions specified by the appropriate ATS authority.

*Note.— This is intended to apply in cases where adequate flight progress data are available from other sources and in other circumstances where the omission of routine reports from selected flights is found to be acceptable.

GENERAL OPERATING PROCEDURES

POSITION REPORTING

Required position reports shall be made to the ATS unit serving the airspace in which the aircraft is operated. In addition, when so prescribed by the appropriate ATS authority in aeronautical information publications or requested by the appropriate ATS unit, the last position report before passing from one FIR or control area to an adjacent FIR or control area shall be made to the ATS unit serving the airspace about to be entered.

If a position report is not received at the expected time, subsequent control shall not be based on the assumption that the estimated time is accurate. Immediate action shall be taken to obtain the report if it is likely to have any bearing on the control of other aircraft.

GENERAL OPERATING PROCEDURES

CONTENTS OF VOICE POSITION REPORTS:

Required position reports shall contain the following elements of information, except that elements (d), (e) and (f) may be omitted from position reports transmitted by radiotelephony, when so prescribed on the basis of regional air navigation agreements:

- a) aircraft identification
- b) position;
- c) time;
- d) flight level or altitude, including passing level and cleared level if not maintaining the cleared level;
- e) next position and time over; and
- f) ensuing significant point.

Note - when assigned a speed to maintain, the flight crew shall include this speed in their position reports.

GENERAL OPERATING PROCEDURES

CHANGE FROM IFR TO VFR FLIGHT

Change from instrument flight rules (IFR) flight to visual flight rules (VFR) flight is only acceptable when a message initiated by the pilot-in-command containing the specific expression “CANCELLING MY IFR FLIGHT”, together with the changes, if any, to be made to the current flight plan, is received by an air traffic services unit. No invitation to change from IFR flight to VFR flight is to be made either directly or by inference.

No reply, other than the acknowledgment “IFR FLIGHT CANCELLED AT ... (time)”, should normally be made by an air traffic services unit.

GENERAL OPERATING PROCEDURES



INTERNATIONAL PHONETIC CHART

A	Alfa	AL-FAH	S	Sirerra	SEE-AIR-RAH
B	Bravo	BRAH-VOH	T	Tango	TANG-GO
C	Charlie	CHAR-LEE	U	Uniform	YOU-NEE-FORM
D	Delta	DELL-TAH	V	Victor	VIK-TAH
E	Echo	ECK-OH	W	Whiskey	WISS-KEY
F	Foxtrot	FOKS-TROT	X	Xray	ECKS-RAY
G	Golf	GOLF	Y	Yankee	YANG-KEY
H	Hotel	HOH-TEL	Z	Zulu	ZOO-LOO
I	India	IN-DEE-AH	1	One	WUN
J	Juliett	JEW-LEE-ETT	2	Two	TOO
K	Kilo	KEY-LOH	3	Three	TREE
L	Lima	LEE-MAH	4	Four	FOW-ER
M	Mike	MIKE	5	Five	FIFE
N	Novemver	NO-VEM-BER	6	Six	SIX
O	Oscar	OS-CAH	7	Seven	SEV-EN
P	Papa	PAH-PAH	8	Eight	AIT
Q	Quebec	KEH-BECK	9	Nine	NIN-ER
R	Romeo	ROW-ME-OH	0	Zero	ZEE-RO

GENERAL OPERATING PROCEDURES

- **READ BACK** is vital for ensuring mutual understanding between the pilot and the controller of the intended plan for that aircraft.
- *The following shall always be read back:*
 - Taxi instructions
 - Level instructions
 - Heading instructions
 - Speed instructions
 - Airways/route clearances
 - Approach clearances
 - Runway in use
 - All clearances affecting any runway
 - Altimeter settings
 - Type of radar services
 - Transition levels

GENERAL OPERATING PROCEDURES

RADIO CHECK

- 1 : Unreadable
- 2 : Readable Now and Then
- 3 : Readable But With Difficulty
- 4 : Readable
- 5 : Perfectly Readable

When your radio requires a radio check, follow this procedure:

1. Call another station and request a radio check.
2. The radio check consists of :

“RADIO CHECK 1,2,3,4,5.

HOW DO YOU READ ME? OVER.”

3. Your call sign should be transmitted during test transmissions.
4. Radio checks should not last more than 10 seconds.

CALL SIGNS FOR AERONAUTICAL STATIONS

<i>Unit or service</i>	<i>Call sign suffix</i>
Area control centre	CONTROL
Radar (in general)	RADAR
Approach control	APPROACH
Approach control radar arrivals	ARRIVAL
Approach control radar departures	DEPARTURE
Aerodrome control	TOWER
Surface movement control	GROUND
Clearance delivery	DELIVERY
Precision approach radar	PRECISION
Direction-finding station	HOMER
Flight information service	INFORMATION
Apron control	APRON
Company dispatch	DISPATCH
Aeronautical station	RADIO

DIFFERENT COMMUNICATION FACILITIES

Aerodrome Control Tower

- A unit established to provide **air traffic control** service for aerodrome traffic. **Ground control** is co-located in this unit which is responsible for the separation of traffic moving on the airport surface.

DIFFERENT COMMUNICATION FACILITIES

Aerodrome Control Tower

Approach Control Facility

- An approach control facility is a terminal ATC facility that provides approach control service in the terminal area.
- Services are provided for **arriving** and **departing** VFR and IFR aircraft and, on occasion, enroute aircraft. In addition, for airports with parallel runways with ILS or LDA approaches, the approach control facility provides monitoring of the approaches.

DIFFERENT COMMUNICATION FACILITIES

Aerodrome Control Tower

Approach Control Center

Area Control Center

- An Area Control Center (ACC), also known as a **Center**, is a facility responsible for controlling *instrument flight rules* aircraft *enroute* in a particular volume of airspace (a *Flight Information Region*) at high altitudes between airport approaches and departures.
- In the United States, such a Center is referred to as an **Air Route Traffic Control Center (ARTCC)**.

DIFFERENT COMMUNICATION FACILITIES

Aerodrome Control Tower

Approach Control Center

Area Control Center

Automatic Terminal Information Service (ATIS)

- The automatic provision of current, routine information to arriving and departing aircraft throughout 24 hours or a specified portion thereof.

DIFFERENT COMMUNICATION FACILITIES

Aerodrome Control Tower

Approach Control Center

Area Control Center

Automatic Terminal Information Service (ATIS)

Flight Service Station

- Is an air traffic facility that provides information and services to aircraft pilots before, during, and after flights, but unlike air traffic control (ATC), is not responsible for giving instructions or clearances or providing separation.
- The people who communicate with pilots from an FSS are referred to as flight service specialists.

INSTRUMENT FLIGHT SEQUENCE

PRE-FLIGHT PHASE:

METAR (current day/latest)

NOTAMS (current day/latest)

- Aerodrome of origin
- Aerodrome of destination
- Alternate aerodrome

WEATHER (current day/latest)

- Aerodrome of origin
- Aerodrome of destination
- Alternate aerodrome
- Enroute

MEL and CDL (*Minimum Equipment List and Configuration Deviation List*)

FLIGHT PLAN

- Composite Flight Plan
- Zulu = VFR to IFR
- Yankee = IFR to VFR

FLIGHT SEQUENCE:

DEPARTURE PHASE

- ATIS/Aerodrome Information
- Get clearance to destination
- Engine Start up/Run up
- Taxi Instructions
- Take off - SID

ENROUTE PHASE

- Position Report
- Handovers – E.g: Approach to Control / Control to Approach

APPROACH PHASE

- ATIS (if avail)
- Initial Contact
- Approach (STAR)
- Landing
- Taxi Instructions
- Parking Bay/Location

PRE-FLIGHT DEPARTURE BRIEFING:

ATIS or Latest Aerodrome Information

- **Type of Aircraft** – C172
- **Status** - (MEL/CDL considerations)
- **NOTAMS** - (*significant - discuss/no significant)
- **Latest Weather Information** – from ATIS
- **General description of the Weather** - (Fine, Fair, Cloudy, etc)
- **Runway Condition** - (Dry, Wet, etc.)
- **Starting sequence** - (Normal/Abnormal)
- **Position** - (Ramp/Bay, etc)
- **Pushback** - (if needed) C172 not needed
- **Expected Taxi routing** - (Ground Chart)
- **Block fuel** – Total fuel to destination + Fuel to alternate + 45mins reserved + 1.1 taxi and takeoff
- **Chart Briefing**
 - Runway in use
 - SID (up to First Waypoint)
 - Total Distance of Flight

Radio Sequence - Departure

1. CLEARANCE DELIVERY

- ATC clearance to (destination)
- Squawk Code
- Requested Altitude or FL

2. GROUND

- Request for Engine start-up.
- Taxi instructions.

3. TOWER

- Contact if Holding short and Runway instructions.
- Will handover by ground for contact when cleared from taxiway traffic.
- Report position (ex: On Taxiway Delta, Holding short C3)
- Departure Clearance
- Report when airborne – Will handover to Approach

4. DEPARTURE / APPROACH

- Report after handover by Tower for further instructions
- When reporting position – Ident, DME and/or Estimate to next waypoint in Zulu time, ALT

Departure Clearance

Procedures for Departing Aircraft:

Standard clearances for departing aircraft contain the following items:

- a) aircraft identification;
- b) clearance limit, normally destination aerodrome;
- c) designator of the assigned SID, if applicable;
- d) cleared level;
- e) allocated SSR code;
- f) any other necessary instructions or information not contained in the SID description, e.g. instructions relating to change of frequency.

Clearances on a SID

Clearances to aircraft on a SID with remaining published level and/or speed restrictions shall indicate if such restrictions are to be followed or are cancelled.

Example: CLIMB VIA SID TO (level):

- i. climb to the cleared level and comply with published level restrictions;
- ii. follow the lateral profile of the SID; and
- iii. comply with published speed restrictions or ATC-issued speed control instructions as applicable.

Radio Sequence – Enroute

APPROACH

- Handover to Approach/Control
- Provides vectors/traffic info if needed
- Position Report – Ident, Distance to next waypoint and/or Estimate to next waypoint (Zulu time), Altitude
- Initial Contact if handed over – Ident, A/C type, Type of flight, DME and/or Time to next waypoint, Altitude, Destination

Radio Sequence – Arrival

1. APPROACH

- Handover to Approach
- Aerodrome Information – From ATIS/Approach
- Initial Contact
- Approach Clearance
- Handover to tower – Usually on Final Approach

2. TOWER

- Report upon handover from Approach.
- Landing Clearance.
- Runway evacuation instructions after landing

3. GROUND

- Taxi Instructions to parking Area

Arrival Clearance

Procedures for Arriving Aircraft:

Standard clearances for arriving aircraft shall contain the following items:

- a) aircraft identification;
- b) designator of the assigned STAR if applicable;
- c) runway- in-use, except when part of the STAR description;
- d) cleared level, and
- e) any other necessary instructions or information not contained in the STAR description, e.g. change of communications.

Clearances on a STAR

Clearances to aircraft on a STAR with remaining published level and/or speed restrictions shall indicate if such restrictions are to be followed or are cancelled.

Example: DESCEND VIA STAR TO (level):

- i. descend to the cleared level and comply with published level restrictions;
- ii. follow the lateral profile of the STAR; and
- iii. comply with published speed restrictions or ATC-issued speed control instructions as applicable.

Instrument Flight Sequence Radio Communications Sample

CLEARANCE DELIVERY:

- Clearance Delivery, good morning/ afternoon
- “Good Morning, go ahead”
- (*Call Sign*), C172, Information (*ATIS*), Request clearance to (*Destination*) @ (*ALT*).
- “Cleared to (*Destination*) via Waypoints and/or Airway, (*ALT*), Squawk (*Code*). Further instructions such as ‘expect radar vectors’.”
- *Read back ATC Clearance – Must be in full, best to prepare paper & pen.
- “*Read back correct, contact* Ground (*Frequency*).”

GROUND:

- (Station) Ground, (Call sign), Good Morning/ Afternoon
- “Good Morning, go ahead”
- (*Call Sign*), on (position), request engine start-up
- “(*Call Sign*), Start-up approved, report when ready to taxi”

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Ground (Taxiing):

- (Station) Ground, (Call Sign) Request taxi instructions to (Runway-in-use)
- “Taxi to (*e.g Hold short runway 13 via F1B*)
- Read back
- (When cleared from taxiway traffic): (Call sign) contact tower (*Tower frequency*).”
- Read back

TOWER:

- Tower, (Call sign), Good Morning,
- “Good Morning, go ahead”
- Tower, (*Call Sign*), position.
- “Line-up and wait (*Runway*)”
- Take off clearance
- Read back
- Report airborne, passing ALT
- Contact departure (*Approach frequency*)
- Read back

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DEPARTURE:

- Departure, (*Call sign*), good morning
- “Good Morning, go ahead”
- *Departure, (call sign), airborne(runway), (Passing ALT), to (Assigned ALT).*
- “Radar contact. Climb and maintain (ALT). Report at (Waypoint) or further instructions, (possible) traffic information.”
- Read back instructions, “Copy traffic, will be on the
Lookout (if negative visual), we have visual on traffic (if visual.)”

DEPARTURE (If handover to next Aerodrome/TMA):

- “Ident, contact (Example) Clark Approach (Frequency), good day.”
- *Contact Control at (Frequency), good day*

DEPARTURE (If handover to Control):

- “Ident, contact Control (Frequency), good day.”
- *Contact Control at (Frequency), good day.*

Control:

- *Control, (Callsign), current position or estimate to nearest fix, (ALT), Estimate to (waypoint) in ZULU time.*
- “(Ident), Radar Contact. Maintain/Climb (ALT), Report at (waypoint),

Traffic information.”

- Read back

Approach:

- *(Example), Clark Approach, Greetings.*
- *"Ident, good morning/afternoon, go ahead."*
- *(Ident), over fix or distance to nearest fix, (ALT), estimate to next waypoint in ZULU TIME) . Request landing instructions/aerodrome information.*
- *"Radar Contact. Cleared to (station VOR if applicable) . Maintain or descend ALT. Expected approach type and runway in use. QNH. Report over (next waypoint/fix). Traffic information"*
- Read back

Destination Aerodrome (when transferred by approach):

- *(Example), Clark Tower, Greetings.*
- *"Ident, good morning/afternoon, go ahead."*
- *(Ident), current position. Request landing instructions*
- *"Ident, Expect RWY in use, QNH Settings, report over next waypoint.*

Maintain/descend (ALT)."

- *Read back.*

When commencing Approach:

- “Ident, cleared for (approach) report at IAF, commencing (approach).”
- *Read back.*

Holding:

- “Ident, Hold over (waypoint/VOR), (ALT).”
- *Read back.*

If cleared for approach:

- “Ident, cleared for (approach) and report at IAF, commencing.”
- *Read back.*
- *Report base/procedure turn or FAP/FAF*
- *Read back*
- *Reporting at base/procedure turn or FAP/FAF*
- “Ident, switch to tower (Frequency).”
- *Read back*

Tower:

- *(Example), Clark Tower, Greetings, on (distance) finals RWY-in-use.*
- “Ident, greetings, cleared to land (RWY in use), Wind Velocity, QNH Settings.”
- *Cleared to land (RWY in use), QNH Setting. No need to read back wind info.*

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COMMUNICATION FAILURE

- When an aircraft station fails to establish contact with the aeronautical station on the designated frequency, it shall attempt to establish contact on another frequency appropriate to the route. If this attempt to fails, the aircraft shall attempt to establish communication with other aircraft or other aeronautical stations on frequencies appropriate to the route
- When an aircraft is **unable** to establish communications due to receiver failure, it shall transmit reports at the scheduled times, or position, on the frequency in use, preceded by the phrase “*TRANSMITTING BLIND DUE TO RECEIVER FAILURE*”.



COMMUNICATION FAILURE

- When an aircraft is **unable** to establish communication due to airborne equipment failure, it shall, if so equipped, **squawk 7600**.
- The aircraft shall transmit the intended message, following this by a complete repetition. During this procedure, the aircraft shall also **advised** the *time of its next intended transmission*.
- When an aeronautical station has been unable to establish contact with an aircraft after calls on the frequencies on which the aircraft is believed to be listening, it shall:
 - Request other aeronautical stations to render assistance by calling the aircraft and relaying traffic, if necessary; and/or
 - Request aircraft on the route to attempt to establish communication with the aircraft and relay messages, if necessary.

DISTRESS AND URGENCY PROCEDURES

- Distress and Urgency conditions are defined as:
 - **Distress:** a condition of being threatened by serious and/or imminent danger and of requiring immediate assistance.
 - **Urgency:** a condition concerning the safety of an aircraft or other vehicle, or of some person on board or within sight, but which does not require immediate assistance.
- Distress messages have priority over all other transmissions, and urgency messages have priority over all transmissions except distress messages.
- Pilots should ask assistance whenever there is any doubt as to the safety of a flight. In this way, the risk of a more serious situation developing can be avoided.

DISTRESS AND URGENCY PROCEDURES

- A distress message should contain as many as possible of the following elements, and, if possible, in order shown:
 - a) Name of the station addressed;
 - b) Identification of the aircraft;
 - c) Nature of distress condition;
 - d) Intention of the person in command;
 - e) Position, level and heading of the aircraft; and
 - f) Any other useful information.

EMERGENCY FREQUENCIES

121.5 Mhz	All military towers, most civil ATS facilities, FSS, VDF, UDF station OSVs
2182 Khz	Some ships and boats at sea, most Coast Guard radio station, most FSS, most commercial coast stations
500 Khz	Most large ships at sea, most Coast Guard radio stations, most FSS, most commercial coast stations
8364 Khz	USN DF Stations, OSVs, most Coast Guard radio station, some FSS

SQUAWK CODES

7700 – Emergency code. It tells the controllers that you have a problem.

7600 – Lost Communication.

7500 – Hijack squawk.

0000 – Military intercept code.

1200 – VFR Squawk Code



EMERGENCY MESSAGE PROCEDURE

1. MAYDAY, MAYDAY, MAYDAY (distress) or PAN, PAN, PAN (uncertainty or alert). SOS (distress) or XXX (uncertainty or alert).
2. Aircraft identification
3. Type of aircraft
4. Position or estimated position and time
5. Heading (true or magnetic)
6. True airspeed or estimated true airspeed
7. Altitude
8. Fuel remaining in hours and minutes
9. Nature of distress
10. pilot's intentions (bailout, ditch, etc.)
11. Assistance desired (fix, steer, escort, etc.)
12. Two 10-second dashes with mike (voice) or key (CW) (voice) or K (CW)

INTERNATIONAL EMERGENCY SIGNALS

- **Urgent** “PAN – PAN – PAN” X X X
- **Safety** “Security-Security-Security” T T T
- **Distress** “Mayday-Mayday-Mayday” S O S

4 C's in EMERGENCY SITUATION

1. **CONFESS**

Confess your predicament to any ground station, you should not wait too long, give RCC a chance.

2. **COMMUNICATE**

Communicate with the ground station and pass on as much of the message on the first transmission. Ground stations need information for best search and rescue operation.

3. **CLIMB**

Climb to a higher altitude, if possible, to get better Directional-finding detection.

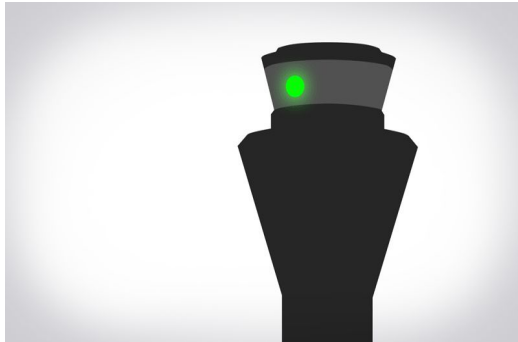
4. **COMPLY**

Comply especially with the advice and instructions received. And to the 4 's stated, add a fifth – **CONSERVE FUEL** by using the most economical power setting and above all, keep **CALM** and **COOL**!

LIGHT GUN SIGNALS



- Light gun signals are used by Air Traffic Control (ATC) towers to direct aircraft if radio communications are out on the ground, if the aircraft has no radio, or communication can't be established.
- If you can't establish radio contact with the tower, circle the field and wait for light gun signals.



ATC Light Signals

GROUND

SIGNAL

AIR

Cleared for Takeoff



Cleared to Land

Cleared to Taxi



Return for Landing

STOP



Give Way
Continue Circling

Taxi Clear of
Runway



Airport Unsafe
DO NOT LAND

Return to Starting
Point on Airport



Not Applicable

Exercise
EXTREME
CAUTION



Exercise
EXTREME
CAUTION

PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES

- **WHAT IS VHF COMMUNICATION?**

VHF, in full very high frequency, conventionally defined portion of the electromagnetic spectrum including any radiation with a wavelength between 1 and 10 metres and a frequency between 300 and 30 Mhz. VHF signals are widely employed for television and radio transmissions.

- **WHAT IS VHF IN AVIATION?**

The type of aircraft communication that most frequently comes to mind is that which takes place between Pilot and Controller. **VHF** communication system are the most widely used for maintaining contact between ground and aircraft. Example of this is **VOR (Very High Frequency Omnidirectional Range)**.

PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES

Frequency Band		Frequency Coverage
Very Low Frequency	VLF	<30 KHz
Low Frequency	LF	30 – 300 KHz
Medium Frequency	MF	300 KHz – 3 MHz
High Frequency	HF	3 – 30 MHz
Very High Frequency	VHF	30 – 300 MHz
Ultra High Frequency	UHF	300 MHz -3 GHz
Super High Frequency	SHF	3 – 30 GHz
Extremely High Frequency	EHF	30 – 300 GHz

PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES

- **HOW DOES VHF COMMUNICATION WORK?**

The primary mode of propagation for VHF radio waves is **through direct waves**. These travel in a straight line, becoming weaker as distance increases. Transmitting and receiving antennas must be able to see each other for communication to be effective, so antenna height is critical in determining range.

- **VHF ALLOCATION**

The range of frequencies allocated to aviation is 117.975 Mhz to 137.00 Mhz. The separation between consecutive frequencies is 8.33 Khz.

PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES

- **VHF** is a line of sight communication system and can be affected by the following:
 - Transmission power of both the aircraft and ground station
 - The height of the transmitter and the receiver
 - Obstacles at or near the transmission site will block or scatter the signals
 - Any obstruction in the light of sight between the aircraft and the ground station will block or scatter signals
 - In certain circumstances both a direct and ground reflected wave may be received which can cause fading
- Because the signals are used over such a short range there is negligible problem from static.

International Morse Code

1. The length of a dot is one unit.
2. A dash is three units.
3. The space between parts of the same letter is one unit.
4. The space between letters is three units.
5. The space between words is seven units.

A ● —
B — ● ● ●
C — ● — ●
D — ● ●
E ●
F ● ● — ●
G — — ●
H ● ● ● ●
I ● ●
J ● — — —
K — ● —
L ● — ● ●
M — —
N — ●
O — — —
P ● — — ●
Q — — ● —
R ● — ●
S ● ● ●
T —

U ● ● —
V ● ● ● —
W ● — —
X — ● ● —
Y — ● — —
Z — — ● ●

1 ● — — —
2 ● ● — — —
3 ● ● ● — —
4 ● ● ● ● —
5 ● ● ● ● ●
6 — ● ● ● ●
7 — — ● ● ●
8 — — — ● ●
9 — — — — ●
0 — — — — —

END OF LESSON